Honors Calculus III, Homework 4

Please upload solutions to **Questions 3a, 3b, 5c and 7** by **8pm on Friday, April 21**. You can use any results you want from lectures, provided you state them clearly, but make sure to write complete proofs. The remaining problems will not be graded, and you should not hand them in, but you are still expected to learn how to do them, unless it says "optional". As usual, for full credit you need to show and explain your work.

1. Let $X \subset \mathbf{R}^n$ and let $f : X \to \mathbf{R}^m$ be a function. Let $x \in X$. Prove that f is continuous at x if and only if $\{f(x_n)\}$ converges to $f(x)$ whenever $\{x_n\}$ is a sequence in X that converges to $x \in X$. (Hint: copy the ϵ, δ-proof from Math 16200 using norms instead of absolute values.)

2. Let $\{x_i\}_{i=1}^{\infty}$ be a sequence in $\mathbf{R}^n$. Prove the following theorems by reducing to the one-dimensional case, which was treated in Math 16200.

   (a) (Bolzano–Weierstrass theorem in $\mathbf{R}^n$.) Assume that $\{x_i\}_{i=1}^{\infty} \subset \mathbf{R}^n$ is bounded: this means that there exists $M > 0$ such that $||x_i|| \leq M$ for all $i \geq 0$. Prove that x_i has a convergent subsequence.

   (b) (Cauchy criterion in $\mathbf{R}^n$.) Assume that $\{x_i\}_{i=1}^{\infty} \subset \mathbf{R}^n$ is a Cauchy sequence: this means that for all $\epsilon > 0$ there exists $N > 0$ such that $m, n \geq N$ implies $||x_m - x_n|| < \epsilon$. Prove that x_i converges to some $x \in \mathbf{R}$.

3. We have seen in class that a function $f : \mathbf{R}^n \to \mathbf{R}^m$ is continuous if and only if all the coordinates $f_1, \ldots, f_m : \mathbf{R}^n \to \mathbf{R}$ are continuous. We can also produce one-variable functions $f^{[i]} : \mathbf{R} \to \mathbf{R}^m$ by putting
$$f^{[i]}(x) = f(0, \ldots, 0, x, 0, \ldots, 0)$$
   where the x is in the i-th coordinate. (Geometrically, this means that we are restricting f to the i-th coordinate axis. Feel free to let $n = 2$ in this problem, if it helps.)

   (a, 5pt) Prove that if f is continuous then $f^{[i]}$ is continuous for all $1 \leq i \leq n$.

   (b, 5pt) Prove that the converse to (a) is not true, by constructing a function $f : \mathbf{R}^n \to \mathbf{R}^m$ such that $f^{[i]}$ is continuous for all i but f is not continuous at zero. For full credit you need to prove that your example works. (Hint: there are examples which take on two values only. Question 1 on this homework might be useful.)

   (c, optional) More generally, construct a function $f : \mathbf{R}^2 \to \mathbf{R}$ which is continuous when restricted to every line through the origin, but is not continuous at the origin.

4. We define the sup-norm on $\mathbf{R}^n$ by the formula
$$||x||_\infty = \sup\{|x_1|, \ldots, |x_n|\}$$
   for all $x \in \mathbf{R}^n$.

   (a) Prove that $||x + y||_\infty \leq ||x||_\infty + ||y||_\infty$ for all $x, y \in \mathbf{R}^n$.

   (b) Prove that $||x||_\infty = 0$ if and only if $x = 0$, and that $|\lambda x||_\infty = |\lambda|||x||_\infty$ for all $\lambda \in \mathbf{R}$ and all $x \in \mathbf{R}^n$.

   (c) Prove that there exist $M_1, M_2 > 0$ such that
$$M_1||x||_\infty \leq ||x|| \leq M_2||x||_\infty$$
   for all $x \in \mathbf{R}^n$.

(Optional) The first two points mean that $||x||_\infty$ satisfies the same properties as $||x||$: every function $\mathbf{R}^n \to \mathbf{R}$ with these properties is called a *norm*. The third point is what it means for the norms $||x||_\infty$ and $||x||$ to be *equivalent*.

5. Let $x \in \mathbf{R}^n$ and choose $\delta > 0$. The open ball of centre x and radius δ is defined to be

$$B(x, \delta) = \{z \in \mathbf{R}^n : ||z - x|| < \delta\}.$$

Similarly, the closed ball of centre x and radius δ is defined to be

$$\overline{B}(x, \delta) = \{z \in \mathbf{R}^n : ||z - x|| \leq \delta\}.$$

(a) Prove that $B(x, \delta)$ is open. (You need to prove that for all $y \in B(x, \delta)$ there exists an open rectangle in $\mathbf{R}^n$ containing y and contained in $B(x, \delta)$.)

(b) Prove that $\overline{B}(x, \delta)$ is closed.

(c, 5pt) Let $U \subset \mathbf{R}^n$. Prove that U is open if and only if the following is true: for all $x \in U$ there exists $\delta > 0$ such that $B(x, \delta) \subseteq U$.

6. Determine whether each of the following sets is open, or closed, or neither.

   (a) $\{(x, y, z) \in \mathbf{R}^3 : x^2 + 2y^2 + 3z^2 \leq 3\}$.
   (b) $\mathbf{R}^2 \setminus \{(0, 0)\}$.
   (c) $\mathbf{R}^3 \setminus \{(x, y, z) \in \mathbf{R}^3 : x + y + z = 0\}$.
   (d) $\mathbf{R}^2 \setminus \{(x, y) \in \mathbf{R}^2 : y = 0, |x| \neq 0\}$.
   (e) The following intersection in $\mathbf{R}$: $\bigcap_{n>0}(-1/n, 1/n)$.

7. (5pt) Let X be a closed subset of $\mathbf{R}$ containing every rational number. Prove that $X = \mathbf{R}$.

Solutions. (1) Assume that f is continuous and $x_n \to x$. Fix $\epsilon > 0$. Then there exists $\delta > 0$ such that $||y - x|| < \delta$ and $y \in X$ implies $||f(y) - f(x)|| < \epsilon$. Since $x_n \to x$ we have $||x_n - x|| < \delta$ for n large enough, and so $||f(x_n) - f(x)|| < \epsilon$ for n large enough. This implies $f(x_n) \to f(x)$.

Conversely, assume that f is not continuous. We need to construct a sequence $x_n \to x$ in X such that $f(x_n)$ does not converge to $f(x)$. Since f is not continuous at x, there exists some $\epsilon > 0$ with the following property: for all $n > 0$ there exists $x_n \in X$ such that

$$||x_n - x|| < \frac{1}{n} \text{ but } ||f(x_n) - f(x)|| \geq \epsilon.$$

By definition $\{x_n\}$ is a sequence converging to x, but since $||f(x_n) - f(x)|| \geq \epsilon$ for all n, we see that $f(x_n)$ does not converge to $f(x)$.

(3a) This can be checked with an explicit ϵ, δ-argument using norms; we can give a quicker argument as follows. Let $g_i(x) = (0, \ldots, 0, x, 0, \ldots, 0)$ be the inclusion of the i-th coordinate axis. By definition, we have $f^{[i]} = f \circ g_i$. Since the composition of continuous functions is continuous, and we are assuming f to be continuous, it suffices to prove that g_i is continuous. But g_i is a linear function, hence it is continuous.

(3b) Let $f : \mathbf{R}^n \to \mathbf{R}$ be the indicator function of the union of the coordinate axes: this means that $f(x) = 1$ if x has at least one zero coordinate, and $f(x) = 0$ otherwise. Then $f^{[i]}$ is the constant function with value 1 for all i, since $f^{[i]}$ is the restriction of f to the i-th coordinate axis. Hence $f^{[i]}$ is continuous for all i.

However, f is not continuous. To see this, let $x_n = (1/n, 1/n, \ldots, 1/n)$ be the vector all of whose coordinates are equal to $1/n$. Then x_n converges to 0, but $f(x_n) = 0$ for all n and $f(0) = 1$, so $f(x_n)$ does not converge to $f(0)$. So f is not continuous at 0 by Question 1.

(5c) Assume that for all $x \in U$ there exists $\delta_x > 0$ such that $B(x, \delta_x) \subset U$. Then

$$U = \bigcup_{x \in U} B(x, \delta_x)$$

since the right-hand side contains all points of U. By part (a), the right-hand side is a union of open sets, hence it is open. So U is open.

To prove the converse, it suffices to prove that if A is an open rectangle, and $x \in A$, then there exists $\delta > 0$ such that $B(x, \delta) \subset A$. Write $A = (\alpha_1, \beta_1) \times \cdots \times (\alpha_n, \beta_n)$. Since $x \in A$, we have $x_i \in (\alpha_i, \beta_i)$ for all i. Since (α_i, β_i) is an open interval, for all i there exists $\delta_i > 0$ such that $(x_i - \delta, x_i + \delta) \subset (\alpha_i, \beta_i)$. Let $\delta = \min_i\{\delta_i\}$.

Then the following smaller open rectangle B (really, a square) is contained in A:

$$B = \{(t_1, \ldots, t_n) \in \mathbf{R}^n : |x_i - t_i| < \delta \text{ for all } 1 \leq i \leq n\} \subseteq A.$$

But B has an alternative definition: by definition of the sup-norm, we find that

$$B = \{t \in \mathbf{R}^n : ||x - t||_\infty < \delta\}.$$

Since

$$||x - t||_\infty \leq ||x - t||$$

for all t, we conclude that $B(x, \delta) \subset B \subset A$.

(7) Since A is closed, if $x \in \mathbf{R}$ is the limit of a sequence x_n of rational numbers, then $x \in A$. But for all $x \in \mathbf{R}$ there exists a sequence of rational numbers converging to x, hence every $x \in \mathbf{R}$ is contained in A, and so $A = \mathbf{R}$.

To see that such a sequence exists, we use the fact from Math 16100 that for all $n > 0$ there exists a rational number

$$x_n \in (x - 1/n, x + 1/n).$$

Then the sequence $\{x_n\}$ is a sequence of rational numbers converging to x.